YTL HACKATHON 2026 · ENTERPRISE TRACK A · BUSINESS CASE

Project Udyam Digital

Making small businesses go digital in one sitting

Accel Business loses 93 of every 100 small-business leads. Quotes take days, KYC is paper-heavy and every product is billed separately. We replace that with one phone-first journey: advisor, instant quote, near-instant KYC and a live order tracker. A rules engine guarantees every quote is valid.

Lead to activation

7.4% → 10.5%

Average days to activate

17.6 → ≈10

fibre bundles; 2–5 days for SIM and 5G

Cost per activation

₹15k → ≈₹9k

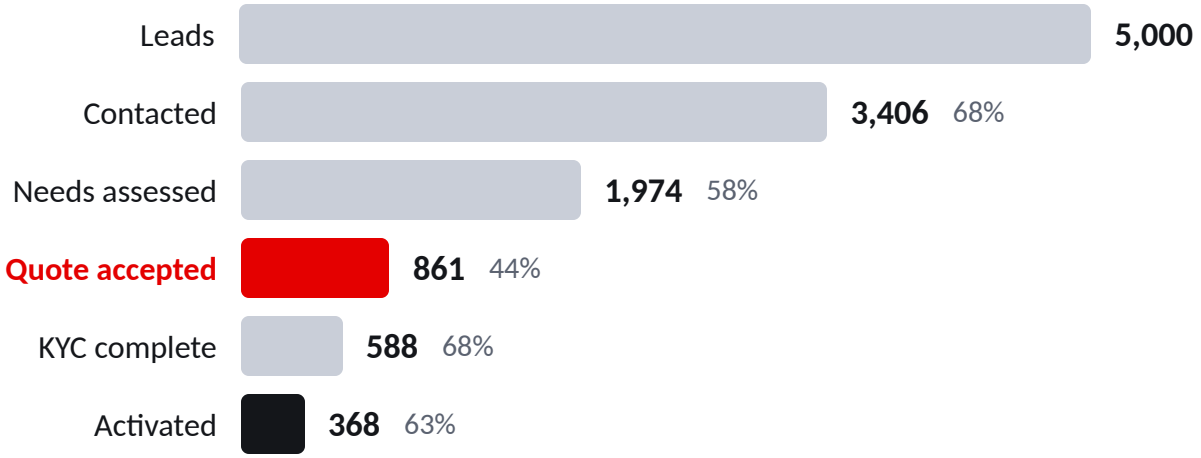
stretch target ₹7,000

Revenue case. +157 activations a quarter adds about **₹10.8 lakh of monthly recurring revenue** at the median bundle (₹6,900), which is a ₹1.3 cr annualised run-rate before churn (about ₹1.1 cr after 12% six-month churn). At the mean bundle (₹12,350) it is ₹19.4 lakh a month, or ₹2.3 cr.

1 · THE FUNNEL

Only 7.4% of leads activate, and every leak has a named cause

5,000 leads, Apr–Jun 2026 (percent shown is of the previous stage)

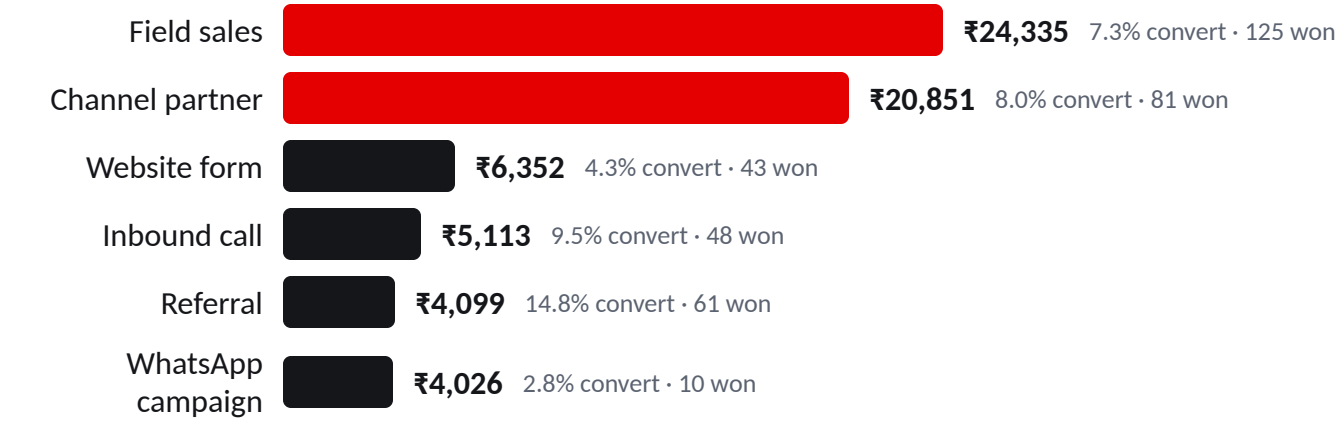


Where the 17.6 days go: about 9 days to reach a quote, 3.7 days from quote to KYC and 4.8 days from KYC to activation. Most of the delay sits before installation, so it can be removed.

STAGE	LOST	WHY (SHARE OF THAT STAGE'S DROPS)	OUR FIX
Lead → contacted	**32%**	50% unreachable after 3 calls · 29% not interested · 20% wrong number	**Reach on WhatsApp first, not by phone**
Contacted → assessed	**42%**	37% had no time for a visit · 21% visit never scheduled	**No visit: the advisor does discovery**
Assessed → quote accepted	**56%**	37% price too high · 19% competitor cheaper · 19% quote too complex · 16% too slow	**Instant one-number quote**
Quote → KYC complete	**32%**	34% missing GST/Udyam doc · 29% paperwork too long · 19% signatory away · 17% mismatch	**Lookup auto-fill, remote e-sign**
KYC → activated	**37%**	45% install delayed over 21 days · 24% cancelled · 19% site not feasible · 13% payment failed	**Feasibility first, live tracker**

Human-led channels cost four to five times more per activation

Cost per activated customer by lead source (total spend ÷ activations)



Caution: the cheap sources are small. WhatsApp produced only 10 activations and referral 61, so their cost may rise as volume grows. We plan on ₹4,000, not less.

9.6% vs 2.0%

activation with a GSTIN versus without. Micro retail without one converts at 1.6%.

25.7% vs 11.1%

six-month churn on bundles up to ₹2,000 a month versus above ₹5,000. Right-sizing the bundle is a retention lever.

14.8% vs 2.8%

referral versus WhatsApp-campaign conversion. Who we pursue matters as much as how.

Prioritise by unit economics, not by lead share

SEGMENT	LEADS	ACTIVATION	COST / ACTIVATION	MEDIAN BUNDLE / MO	6-MO CHURN	SHARE OF ACTIVATED REVENUE	ROLE IN ROLLOUT
Professional offices	25%	11.1%	₹10,695	₹6,575	13.8%	**21%**	**1 · Wedge** Best cost-to-value, high GSTIN coverage
Small multi-site businesses	15%	11.0%	₹15,614	₹28,080	10.8%	**57%**	**2 · Revenue engine** Digital quote and KYC, field-led install
Micro retail & services	54%	4.1%	₹20,007	₹2,470	22.7%	**6%**	**3 · Zero-touch only** Viable only at ≈₹4,000 per activation
Small manufacturers	6%	12.8%	₹15,324	₹15,910	8.1%	**16%**	**4 · Field-led** Keep human this year

Micro works only when it is zero-touch

Micro is 54% of leads but 6% of revenue. At today's ₹20,007 per activation and a ₹2,470 bundle, payback is about 23 months at an assumed 35% margin, against 23% churn. At ₹4,000, the cost of our cheapest observed channel, payback falls to about 4.6 months.

The micro upside, sized honestly

Micro owners with a GSTIN (1,431 leads, 6.3%) reaching the professional-office rate of 12.1% adds about 83 activations, roughly ₹25 lakh a year. Worth doing, not worth overselling. The 35% margin is our assumption; the data has no margin figure.

One journey, four modules, one bill

1

Advisor

Five questions or free text, in the owner's language. Rules engine picks valid bundles; the LLM explains why.

2

Quote & checkout

One monthly price, live add-on toggles, UPI or NetBanking. Valid for 7 days.

3

Smart KYC

Consent first, then GSTIN and Udyam lookup auto-fill, OCR cross-checks and remote e-sign.

4

Tracker

Per-product install timeline, SLA timers and proactive WhatsApp updates.

BUNDLE	FOR	CONTENTS	FROM / MONTH
Digital Dukaan	Micro, no GSTIN needed	5G AirFiber, 2 SIMs, POS, CCTV	**₹2,295**
Office Pro	Clinics, CA, coaching	Fibre 300, email suite, endpoint security, priority SIM (5 users)	**₹3,218**
Branch Command	Multi-site	Fibre 300, SD-WAN, failover, 2 cameras per site	**₹4,896 / site**

Prices are catalogue floors at 24-month terms. GST and one-time setup are shown separately in the quote.

Architecture rule

The rules engine owns city tier, size, GSTIN and dependency checks, so every quote is valid by construction. The LLM only translates and explains. It never sets prices, SKUs or eligibility.

12 specimen applicants tested: 5 pass, 7 stopped correctly

Advisor

Five questions (or free text such as “2 dukaan hain, UPI chalta hai”) fill the profile. The engine filters all 25 products by tier, size, GSTIN and dependencies. The LLM then explains two options, recommended and budget, with a monthly price and an install date. Ineligible items are explained with the nearest valid alternative.

Smart KYC

Consent screen first. GSTIN format and real checksum (4th character X accepted), registry status (Active, Suspended, Cancelled), Udyam match, four-way name match, address proof within 90 days of 2026-07-15 and signatory authority. OCR misreads of O/0 and I/1 are corrected against the registry.

Why it matters: 63% of post-quote drops are missing or slow paperwork, and 19% are an unavailable signatory. Remote e-sign removes that second cause.

Our pipeline run on the specimen documents (not an official answer key)

K01	Cancelled GSTIN, no Udyam number
K02	Passes after e-sign step (O/0 misread fixed)
K03	Auto-approved
K04	GST scan unreadable: re-upload or DigiLocker
K05	No Udyam certificate
K06	GSTIN checksum fails, likely altered
K07	Name differs from registry
K08	Passes after e-sign step (O/0 misread fixed)
K09	Passes after e-sign step
K10	GSTIN checksum fails, likely altered
K11	Address bill 257 days old (limit 90)
K12	Auto-approved

The digital journey feeds the channel instead of competing with it

Tracker

Timeline per product from catalogue install days: SIM 1 day, 5G AirFiber 2, fibre 5–7, SD-WAN 14, leased line 21–30. Auto-escalation at day 14. Owner and partner see the same status. Goal: status calls per customer from 3.0 to under 0.5.

Propensity score

A logistic model on the 5,000 leads (GSTIN, lead source, size, spend). Held-out AUC is 0.72, and the top 20% of leads hold 37% of activations. Field and partners call the top decile first, not cold lists.

Feasibility first

41 leads were lost because fibre was not feasible at the site. An address-level check before the quote stops us selling what we cannot install, and offers 5G AirFiber instead.

JOURNEY TYPE	OWNER	WHY
Micro, GSTIN-free, up to 3 sites, standard products	**Fully digital**	Only viable near ₹4,000 per activation
Professional offices	**Digital plus remote assist**	Best economics, standard needs
Multi-site, leased lines, feasibility cases	**Field or partner**	Site surveys and 21–30 day installs
Leads the advisor could not close	**Partner, from the scored list**	Partner-originated leads stay attributed with incentives intact

Conservative on purpose: 10.5%, not the 14% we modelled

METRIC	BASELINE (APR–JUN 2026)	6-MONTH TARGET	MECHANISM
Lead → activation	7.4% (368 of 5,000)	**10.5% (≈525)**	Instant quote, auto-KYC, feasibility first
Average time to activate	17.6 days	**≈10 days fibre · 2–5 days SIM/5G**	Same-day quote and KYC; installs bound by catalogue days
Cost per activation	≈₹15,000	**≈₹9,000 (stretch ₹7,000)**	Zero-touch micro; partners call the top decile only
Status-chase calls	3.0 per customer	**under 0.5**	Proactive tracker
Micro without GSTIN activation	1.6%	**4% or more**	Digital Dukaan and an assisted GSTIN nudge
Six-month churn	15.2%	**under 12%**	Right-sized bundles

Revenue case

157 extra activations a quarter (525 vs 368), deliberately about half of the ≈340 we modelled. At the median bundle of ₹6,900: **₹10.8 lakh a month**, ₹1.3 cr annualised. At the mean ₹12,350: ₹19.4 lakh a month, ₹2.3 cr. We plan on the median.

Reading the cost target

Spend today is ₹55 lakh a quarter. With micro at ₹4,000 and other channels about 20% cheaper, 525 activations cost about ₹47 lakh, or ≈₹9,000 each. Revenue is gross, before churn. Targets are planning figures for a pilot.

8 · ROLLOUT AND RISKS

Three gates, each with a kill criterion

1 Pilot · weeks 1–6, 2 cities

Professional offices only, website and inbound entry.
Pass: 9% or more activation, 40% or more KYC straight-through, zero invalid quotes.
Fix: any ineligible product quoted, or lookup errors above 2%.

2 Micro zero-touch · weeks 7–14

Digital Dukaan on WhatsApp, with a referral loop.
Pass: cost per activation ₹4,000 or less, install NPS 8 or more.
Fix: above ₹6,000, restrict to referral-sourced micro.

3 Multi-site · weeks 15–24

Digital quote and KYC, field-led install, partner console.
Pass: quote to KYC in 1 day or less (now 3.7), no feasibility surprises after quote.

RISK	MITIGATION
LLM recommends an ineligible product	Rules engine is the only path to a quote; the LLM cannot see prices or SKUs
Fraudulent GSTIN or documents	Checksum, registry status, four-way match, 90-day freshness, human review queue
Install capacity misses the SLA	Feasibility-first quoting; SLA timers surface backlog before day 21
Partners feel bypassed	Attribution and incentives preserved; they get scored leads
Micro churn stays near 23%	Payback holds only at ≈₹4,000; right-size bundles, track churn from month one

Privacy (DPDP Act, 2023): consent before collection, minimisation, purpose limitation, masking, self-serve rights. See the separate one-page note.

Why this wins

Every module maps to a measured leak in the 5,000-lead data.

Rules engine plus LLM: accurate by construction, warm in ten languages.

Honest economics: 10.5% not 14%, about 10 days bounded by install times, revenue shown at median and mean, margin assumption stated.

Channel-positive: scored routing makes partners more productive, with attribution intact.

Compliant by design: consent first, masked identifiers, minimal data.

THE THREE-MINUTE DEMO

1. Hindi-speaking kirana owner answers five questions; advisor quotes Digital Dukaan at ₹2,295 a month with a CCTV toggle
2. Consent screen, then KYC by lookup; OCR catches a stale address bill (K11)
3. Guardrail moment: K01's cancelled GSTIN is stopped at lookup with a guided recovery path
4. Tracker: 5G install in 2 days, no phone calls
5. Partner console: scored leads, "call these ten first"

Close: **7.4% → 10.5% · 17.6 → ≈10 days · ₹15k → ≈₹9k** per activation · **₹1.3 cr** annualised run-rate